

Educational AI Knowledge Distillation Edge Inference

WhyWrong: Mathematics Misconception Diagnosis

An automated diagnostic engine that maps student math errors to an underlying 2,587-class misconception taxonomy, generating targeted Socratic hint ladders and live session insights on lightweight CPU hardware.

2,587

Taxonomy Misconceptions

<400 ms

p50 CPU Latency

264×

Cost Advantage vs Frontier APIs

1 Core

Zero GPU Dependency at Serving

From Binary Scoring to Root-Cause Diagnosis

Conventional tools evaluate whether an answer is correct or incorrect. WhyWrong identifies the underlying misconception rule and guides the student through self-correction.

STANDARD PRACTICE PLATFORMS

Binary Feedback Failure

Student evaluates: $7/12 - 3/12 = 4/24$

✖ "That is not quite right, try again!"

The student repeatedly applies the same flawed procedure (subtracting denominators) without understanding why it fails, leading to frustration and disengagement.

WHYWRONG DIAGNOSTIC ENGINE

Pedagogical Socratic Ladder

Identifies: #172: Subtracts numerators and denominators

🎯 Targeted 3-Step Socratic Progression

- **Step 1 (Guiding Question):** "What parts of the fraction change when parts of the same size are removed?"
- **Step 2 (Conceptual Clue):** "Think about eighths or quarters: if you have 7 eighths and lose 3, the size remains eighths."
- **Step 3 (The Rule):** "When fractions have common denominators, subtract only numerators."

Session Intelligence for Educators

During a 45-minute 1-on-1 tutoring session, educators cannot manually cross-reference dozens of disparate errors. WhyWrong clusters mistakes across different subjects to uncover shared root causes.

1. MULTITOPIC AGGREGATION

Disparate Errors

Student Priya submits incorrect answers across 6 questions spanning 4 separate topics:

- Linear Equations: $4(m - 1) = 20$
- Expanding Single Brackets: $x(3x - 5)$
- Factorising Expressions: $24abc^2 - 16b^2c$
- Algebraic Fractions: $4 \div 2(x + 3)$

2. EMBEDDING CLUSTERING

Root Cause Alert

100% Shared Cause Confidence: 89%

#220: Only multiplies the first term in bracket expansion

Rather than 4 separate knowledge gaps, WhyWrong's embedding space clusters the mistakes into a single conceptual misunderstanding.

3. ACTIONABLE INTERVENTION

Immediate Tutor Tools

Suggested Opening Prompt:

"Can you talk me through how you expanded the second term in this bracket?"

Mastery Check Probes:

- Probe 1: Apply distributive property across terms with mixed signs.
- Probe 2: Factorise negative coefficients.

Two-Stage Retrieval & Distillation Pipeline

Heavy frontier models run offline once for synthetic augmentation and teacher supervision. Serving is distilled into a compact 22M INT8 ONNX model running on commodity CPU.

OFFLINE TRAINING (GPU INFRASTRUCTURE)

Training & Distillation

- **Synthetic Augmentation:** Qwen2.5-7B generated 17,490 diagnostic question pairs covering 1,498 zero-shot taxonomy classes (58% of taxonomy).
- **Retriever Training:** BGE-small (33M) contrastive fine-tuning with hard-negative mining; real pairs oversampled 6× to preserve seen performance.
- **Teacher Model:** Qwen2.5-7B + LoRA listwise cross-encoder reranker trained over top-25 candidate pools.
- **Knowledge Distillation:** Teacher soft scores distilled into a 22M MiniLM-L6 cross-encoder via KL divergence loss.
- **Explanation Caching:** Socratic hint ladders for all 2,587 misconceptions precomputed and indexed into SQLite.

PRODUCTION RUNTIME (1 CPU CORE)

Deterministic Serving Path

1. Bi-Encoder Embedding (INT8 ONNX)

~15 ms

Encodes question and student response into 384-d normalized vector.
2. NumPy Matrix Multiplication

<1 ms

Evaluates query vector against 2,587 × 384 index (3.97 MB) to select top-25.
3. Distilled Cross-Encoder (INT8 ONNX)

~360 ms

Reranks top-25 candidate misconceptions down to top-1 prediction.
4. SQLite Primary-Key Lookup

<1 ms

Fetches Socratic hint progression and educator probes from indexed storage.

Empirical Results on Held-Out Test Split

Evaluated on the NeurIPS 2024 Eedi competition dataset (882 held-out test queries, 64% featuring unseen gold labels).

HELD-OUT METRIC COMPARISON

MODEL / APPROACH	PARAMS	MAP@25	RECALL@25	R@25 (UNSEEN)
Off-the-shelf bge-small	33M	0.153	0.466	0.504
Off-the-shelf bge-m3	568M	0.160	0.478	0.536
Fine-tuned retriever (real data)	33M	0.246	0.643	0.676
+ Synthetic data + 6× oversample	33M	0.266	0.661	0.710
7B LoRA reranker (teacher)	7.6B	0.361	0.662	0.712
WhyWrong Shipped (INT8 ONNX)	22M	0.261	0.650	0.696

COST & HARDWARE COMPARISON

\$0.002

WhyWrong Cost / 1k Queries

\$1.15

Frontier LLM API / 1k Queries

Production Economics (1M queries/mo)

- Frontier API (Claude Sonnet): \$1,153.90 / month + cloud latency & rate limits.
- Frontier API (Claude Haiku): \$576.90 / month + network round trips.
- WhyWrong (Commodity vCPU): \$2.18 / month on a \$5/mo shared VPS.

264× Lower Cost · 100% Data Privacy · Offline Capable

Failure Analysis & Product Roadmap

A credible machine learning submission analyzes where the system struggles. We cataloged held-out failure cases to drive our next architectural iteration.

OBSERVED FAILURE MODES

1. Diagram & Visual Dependency

Questions requiring unrendered geometric diagrams or number-line intervals cause text-only embeddings to misclassify procedural errors.

2. Overlapping Distractor Misconceptions

Certain incorrect options are mathematically consistent with two distinct misconception labels. Forcing a greedy top-1 prediction leads to arbitrary selection.

DEVELOPMENT ROADMAP

1. Interactive Disambiguation Dialogues

When top candidates have overlapping confidence scores, prompt the student with one diagnostic clarifying question instead of guessing.

2. Curriculum-Aware Adaptive Sequencing

Feed live misconception state directly into knowledge-tracing engines to dynamically build customized problem progression paths.

3. Cross-Discipline Taxonomy Scaling

Expand the bounded retrieval architecture to physics problem-solving, grammar rules, and introductory Python programming.